Statistics: **Statistics can be defined as the science of collection, organization, presentation, analysis and interpretation of numerical Data**

Functions:

- It represents facts in a definite form.
- It simplifies mass of figures.
- It facilitates comparison.
- It helps in formulating and testing hypothesis.
- It helps in prediction.
- It helps in the formulation of suitable policies.

Characteristics of Statistics (Easy Version)

- Statistics are based on real facts, not guesses.
- They are always shown in numbers.
- Many different factors can affect the results.
- The numbers are counted or estimated carefully and accurately.
- Data is collected for a specific reason or goal.
- Information is gathered in a planned and organized way.
- The data should be easy to compare with other similar data

Scope of Statistics: In the real world, there are no sectors where Statistics can't be used. Some scopes of' Statistics are given below:

- Statistics if Agricultures.
- Statistics in Planning.
- Statistics in Economics.
- Statistics in Business management.
- Statistics in Industry.
- Statistics in the State.
- Statistics in Biology.
- Statistics in Psychology.
- Statistics in Education.
- Statistics in Medicine etc.

Limitations of Statistics (Easy Version)

- Statistics doesn’t focus on just one number or case.
- It only works with things that can be measured in numbers.
- It doesn’t study one person or thing alone — it looks at groups.
- The results are based on averages, so they might not apply to everyone.
- Data is often collected from samples, so it's just an estimate for the whole group.
- Statistics can be used in the wrong way to mislead people.

Chapter 2

Population: population means an aggregate of elements processing certain characteristic of interest in any particular investigation or inquiry. It is generally named after the characteristics studied. It maybe finite or infinite.

Example: if we are interest to investigate the previous results of a group of students in UITS Rajshahi campus then all the students of that group together comprise the population.

Sample: A sample in the representative part of a population. That is a sample is a set of **n** observations drawn from the population.

Example: results of some students in a group in UITS Rajshahi campus.

Variable: The measurements of elements of a population having certain characteristics may vary from element either in magnitude or in quality. These measurable characteristics are called variable. Generally, variables represented by x, y, z etc.

Constant: if the characteristics of the measurement of element of a population do not change element to element, then these measurable characteristics are called constant. Generally, its denoted by a, b, c etc.

Basic Concepts of Statistics (Easy Version)

Population

A population means the whole group of items or people we want to study. Everyone in this group shares a certain feature we are interested in.

Example: If we want to study the exam results of a group of students at UITS Rajshahi campus, then all those students are the population.

Sample

A sample is a smaller part taken from the population. It represents the whole group.

Example: If we only check the results of some students from that group, those students form a sample.

Variable

A variable is something that can change or vary from one person or item to another in a population. It could be a number (like marks) or a quality (like grade).

Examples of variable symbols: x, y, z

Constant

A constant is something that does not change. It stays the same for every person or item in the population.

Examples of constant symbols: a, b, c

[then pic er math 4 tah]

****Frequency Distribution:** Arrangement of observational data according to frequency of the observations is called frequency distribution.

Using the following steps, we can construct a frequency distribution:

1. Finding the Range
2. Decision about the number of classes
3. Choosing the number of class interval
4. Give Tick marks or tally marks
5. Counting of frequency

Importance of frequency distribution:

1. It shows data in a short and simple way.
2. It helps collect and organize data quickly and with less effort.
3. We can use it to make graphs, charts, and diagrams easily.
4. It helps us find important features like average, spread, shape, etc.
5. It lets us compare two or more sets of data using the same class intervals.

#Example: the following data represent the number of workers in a different small-scale industry in a country.

19	25	28	32	26	25	25	20	20	22	24	26	28	30	35	32
17	20	22	22	24	25	26	28	20	18	26	28	30	30	32	34
31	36	30	35	28	27	21	24	20	18	15	15	15	18	20	22
36	26	21	23	24	26	28	30	32	34	28	27	15	20	19	26
16	24	20	18	20	20	24	27	25	25	25	26	20	21	20	28
17	30	32	33	30	28	26	24	26	24	20	18	19	18	15	16
23	18	15	17	18	20	20	18	18	19	20	21	27	25	26	19
29	20	24	26	27	29	30	32	34	28	30	27	26	28	28	33

Construct a frequency distribution using suitable class interval.

Solution: Here we have,

$N=128$

$X_{(1)} = 15$ and $X_{(n)} = 36$

$R = X_{(n)} - X_{(1)} = 36 - 15 = 21$

$K = 1 + 3.33 \log_{10}(N)$

$= 1 + 3.33 \log_{10}(128)$

$= 8.017$

≈ 8

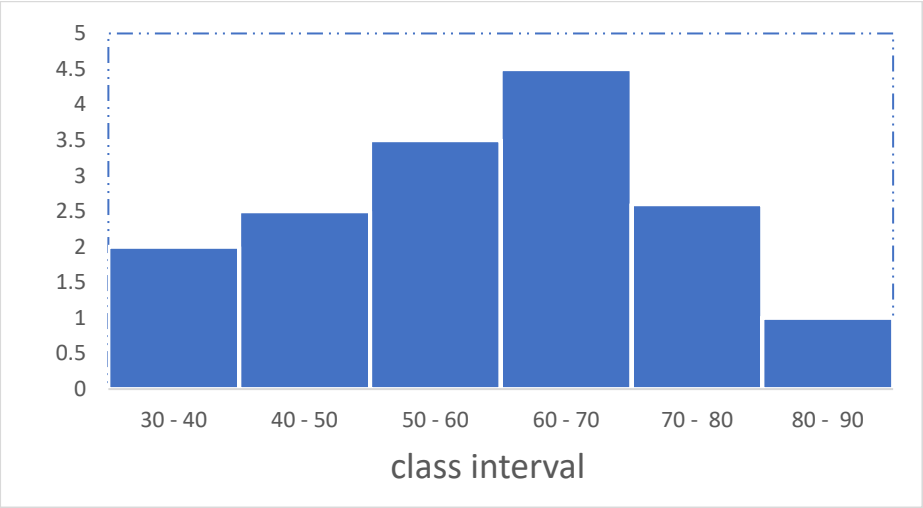
Class interval, $h = \frac{R}{k}$

$= \frac{21}{8} = 2.63 \approx 3$

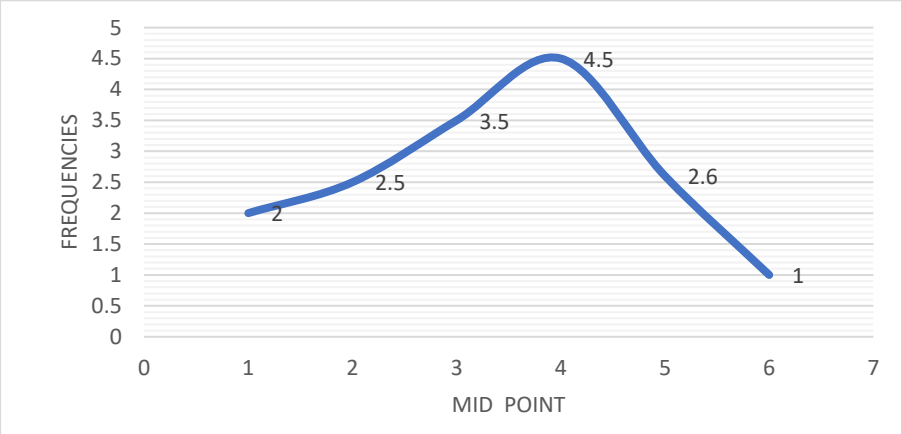
The distribution of industries by number of workers.

Class interval	Mid-value $X_{(1)}$	Tally marks	Frequency $f_{(i)}$	Cumulative frequency (C.F)	
				From top	From bottom
15 - 18	16.5	𝍸 𝍸 ||	12	12	128
18 - 21	19.5	𝍸 𝍸 𝍸 𝍸 𝍸 𝍸 |	31	43	116
21 - 24	22.5	𝍸 𝍸	10	53	85
24 - 27	25.5	𝍸 𝍸 𝍸 𝍸 𝍸 𝍸	30	83	75
27 - 30	31.5	𝍸 𝍸 𝍸 𝍸	20	103	45
30 - 33	31.5	𝍸 𝍸 𝍸 |	16	119	25
33 - 36	34.5	𝍸 ||	7	126	9
36+	37.5	||	2	128	2

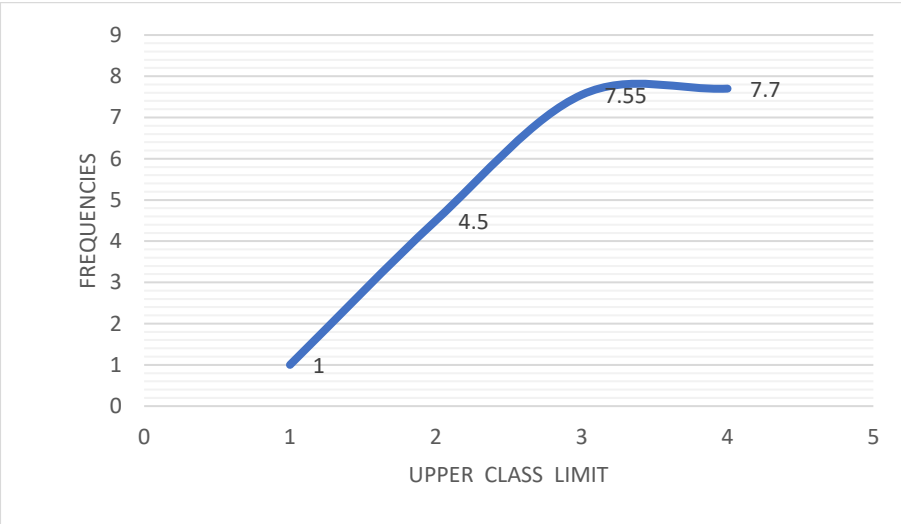
(a) ***Histogram:** To construct a histogram, we put class interval in the X-axis and the frequencies in the Y-axis.



(b) ***Frequency curve:** to construct a frequency curve, we put mid point in the X-axis and the frequencies of the class interval in the Y-axis and add the points secuencely by free hand.



(c) Ogive curve: To construct Ogive curve, er put upper limit of the class interval in the X-axis and the Cumulative frequencies in the Y-axis, Finally add these points by free hand



Probability: Probability is the chance of something happening, measured from 0 to 1. If it's 0, the event can't happen; if it's 1, the event is certain. For example, when rolling a six-sided die

Experiment: It is an act which can be departed under certain identical condition. Example, tossing a coin , throwing a dice ….

Sample space: The collection or totality of all possible outcomes of a random experiment is called sample space. It is usually denoted by **S .**

Example: if we toss a coin, the sample space is.

S = { H , T }, where H = head, T = tail
Of the coin respectively

If a dice is thrown, the sample space is,

S = { 1,2,3,4,5,6 }

Sample point: An element of the sample space is called a sample point. Sample point usually denoted by small letters.

If S or Ω = { 1,2,3,4,5,6 }
Then, 1,2… etc are sample point

Mutually exclusive event: If A and B be two events in the sample space Ω or S, then they are said to be mutually exclusive event if $A \cap B = \emptyset$. That is two events are said to be mutually exclusive if they have no common points.

Example, Let S or Ω = { HH,HT,TH,TT }
And A = { HH }, B = { TT }
Since $A \cap B = \emptyset$, thus A and B are mutually Exclusive events.

#Prove that,

a) $\mathbf{0 \leq P(A) \leq 1}$
b) $\mathbf{P(A) + P(\bar{A}) = 1}$
c) $\mathbf{P(\emptyset) = 0}$

(a) Proof: Let, A be an event from a sample space S, where,

The total number of sample points in S $= n$
The number of favorable sample points in A = m

Thus,

The probability of event A, $P(A) = \frac{m}{n}$

There arises some situation in event A,

i. No favorable outcomes, m=0
ii. All outcomes in favor of A, m=n
iii. Some outcomes in the sample space in A, $0 < m < n$

Thus, the value of **m** is lies from **0** to **n** , that is,

$0 < m < n$

Or, $\frac{0}{n} \leq \frac{m}{n} \leq \frac{n}{n}$

Or, $0 \leq P(A) \leq 1$

[proved]

(b)proof: Let A be any event in the sample space , S

So event **A** and $\bar{A}$ are mutually exclusive and exustive,

Thus,

$$A \cup \bar{A} = S$$

Or, $P(A \cup \bar{A}) = P(S)$

$\therefore P(A) + P(\bar{A}) = 1$ [proved]

(C)proof: Let **S** be a sample space

Thus, $P(S) = 1$

Now, $\emptyset \cup S = S$

Or, $P(\emptyset \cup S) = P(S)$

Or, $P(\emptyset) + P(S) = 1$

Or, $P(\emptyset) + 1 = 1$

$P(\emptyset) = 0$

[proved]

Difference between correlation coefficient and regression coefficient:

Regression	Correlation
Regression means determination of estimated values of the independent variable with respect to one or more independent variables.	Correlation means the linear relationship of two or more dependent variables with each others.
In regression influenced variable is called dependent variable and other variables are independent.	In correlation the variables are dependent on each other variable. There is no concept of independent variables.
In regression, only the dependent variable needs to be random	In correlation, both the variables must be random
In regression, either $y = f(x) or\ x = f(y)$ or both may be true .	In correlation, either $y = f(x)\ and\ x = f(y)$ are to be true .
$b_{xy} \neq b_{yx}$	$r_{xy} \neq r_{yx}$
It has no upper and lower limit	$-1 \leq r \leq +1$
Regression coefficient are independent of change of origin but not of scale	It is independent of both change of origin and scale.